ME/CE/AM 295

Agentic AI in Scientific Discovery

9 units (3-5-1)

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TAs:
Xiaoxiao (Alice) Xiong
Hyoseo (Lauren) Yoon

Welcome!

California Institute of Technology
Division of Engineering and Applied Science

The Dawn of Agentic AI in Engineering



Week 1: From Language Models to Autonomous Agents

TAs intro

Class logistics and more

Homework, Labs and Final projects

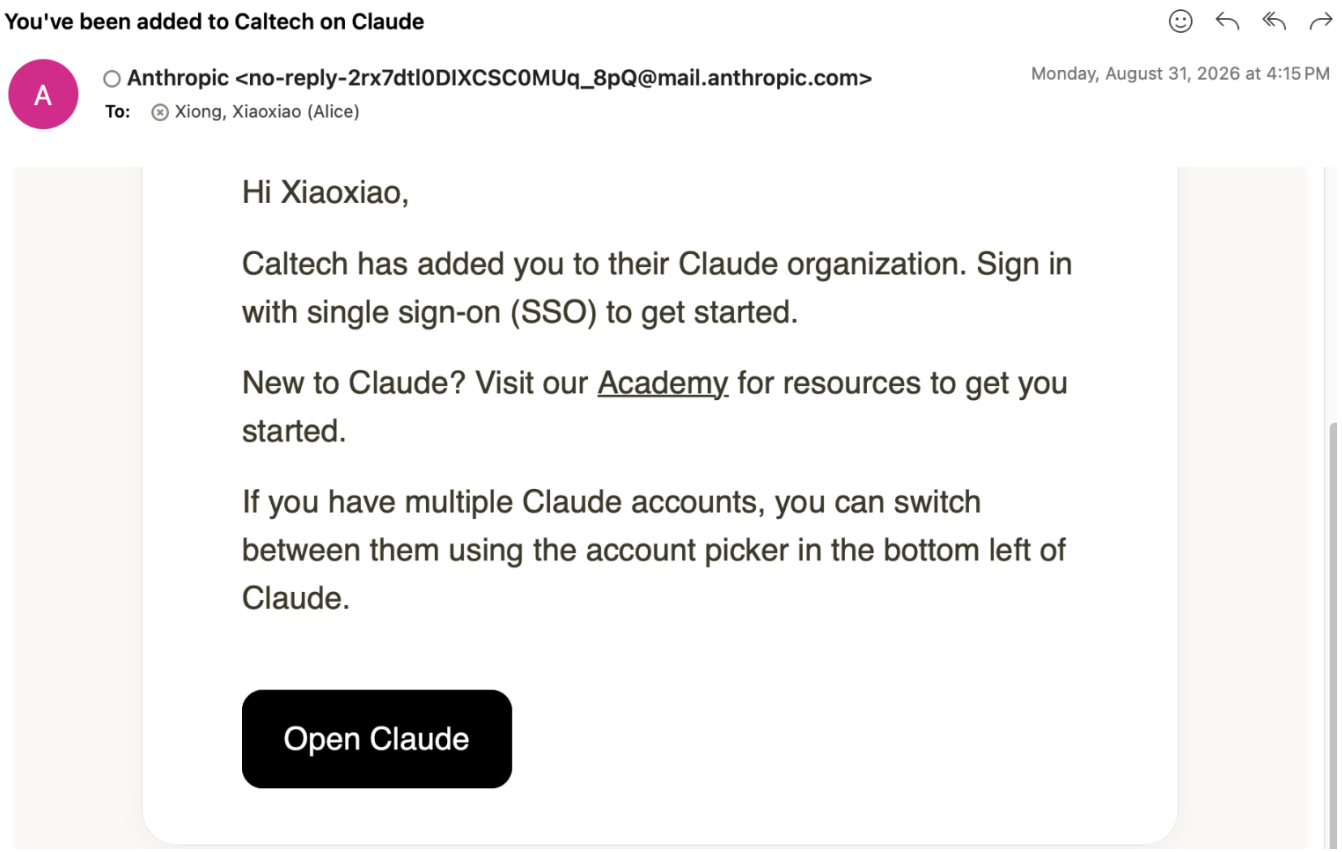
What we provide for this course

1. Claude Code subscription

1. AWS workspace

1. Github space

What we provide for this course – 1. Claude subscription



You've been added to Caltech on Claude

A

Anthropic <no-reply-2rx7dtl0DIXCSC0MUq_8pQ@mail.anthropic.com>

To: Xiong, Xiaoxiao (Alice)

Monday, August 31, 2026 at 4:15 PM

Hi Xiaoxiao,

Caltech has added you to their Claude organization. Sign in with single sign-on (SSO) to get started.

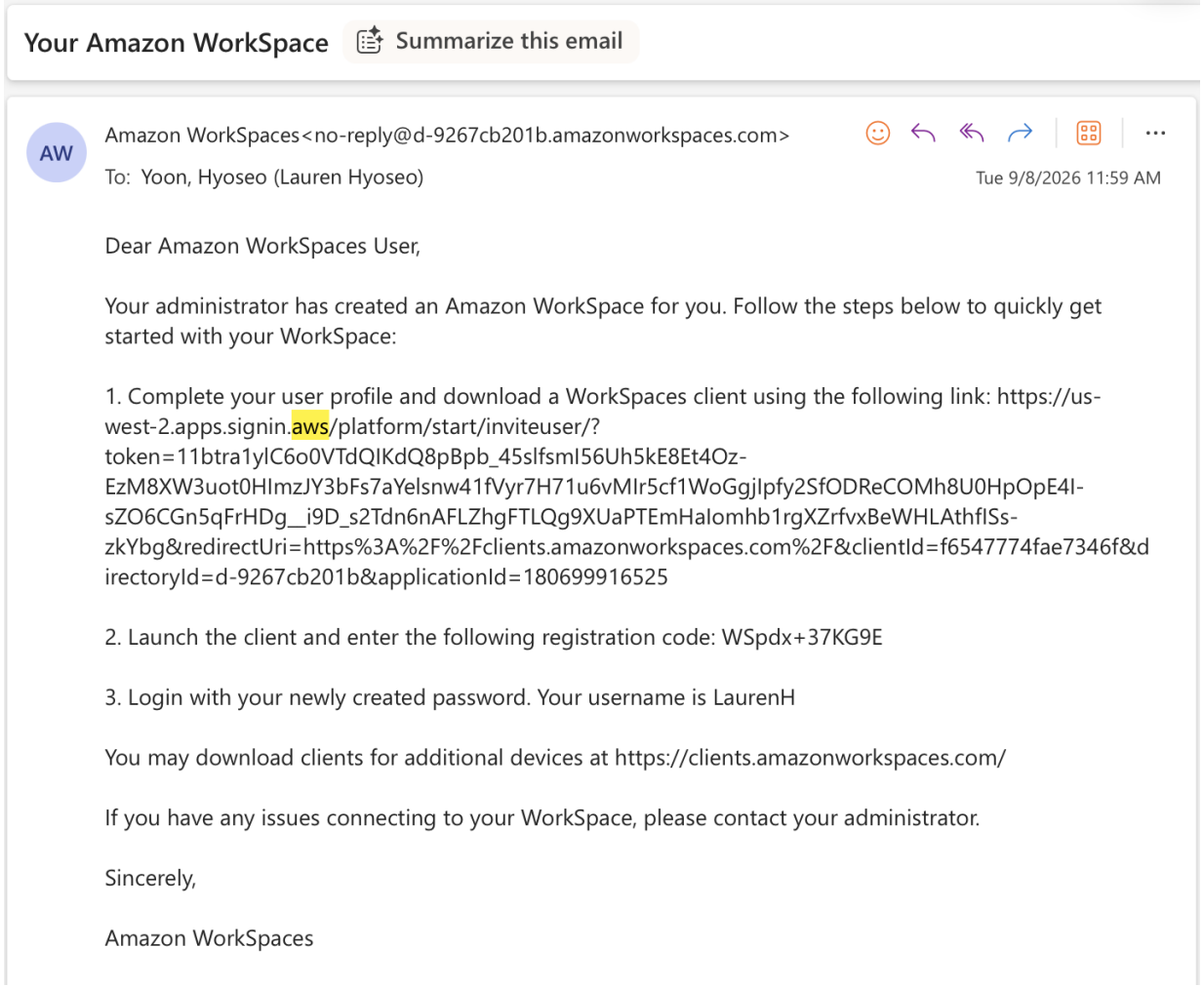
New to Claude? Visit our Academy for resources to get you started.

If you have multiple Claude accounts, you can switch between them using the account picker in the bottom left of Claude.

Open Claude

Claude pro subscription provided through Caltech.

What we provide for this course – 2. AWS workspace



Your Amazon WorkSpace Summarize this email

AW

Amazon WorkSpaces<no-reply@d-9267cb201b.amazonworkspaces.com>

To: Yoon, Hyoseo (Lauren Hyoseo) Tue 9/8/2026 11:59 AM

Dear Amazon WorkSpaces User,

Your administrator has created an Amazon WorkSpace for you. Follow the steps below to quickly get started with your WorkSpace:

1. Complete your user profile and download a WorkSpaces client using the following link: https://us-west-2.apps.signin.aws/platform/start/inviteuser/?token=11btra1ylC6o0VTdQIKdQ8pBpb_45slfsmI56Uh5kE8Et4Oz-EzM8XW3uot0HImzJY3bFs7aYelsnw41fVyr7H71u6vMIr5cf1WoGgjIpfy2SfODReCOMh8U0HpOpE4I-sZO6CGn5qFrHDg__i9D_s2Tdn6nAFLZhgFTLQg9XUaPTEmHaIomhb1rgXZrfvxBeWHLAthfISs-zkYbg&redirectUri=https%3A%2F%2Fclients.amazonworkspaces.com%2F&clientId=f6547774fae7346f&directoryId=d-9267cb201b&applicationId=180699916525

2. Launch the client and enter the following registration code: WSpdx+37KG9E

3. Login with your newly created password. Your username is LaurenH

You may download clients for additional devices at https://clients.amazonworkspaces.com/

If you have any issues connecting to your WorkSpace, please contact your administrator.

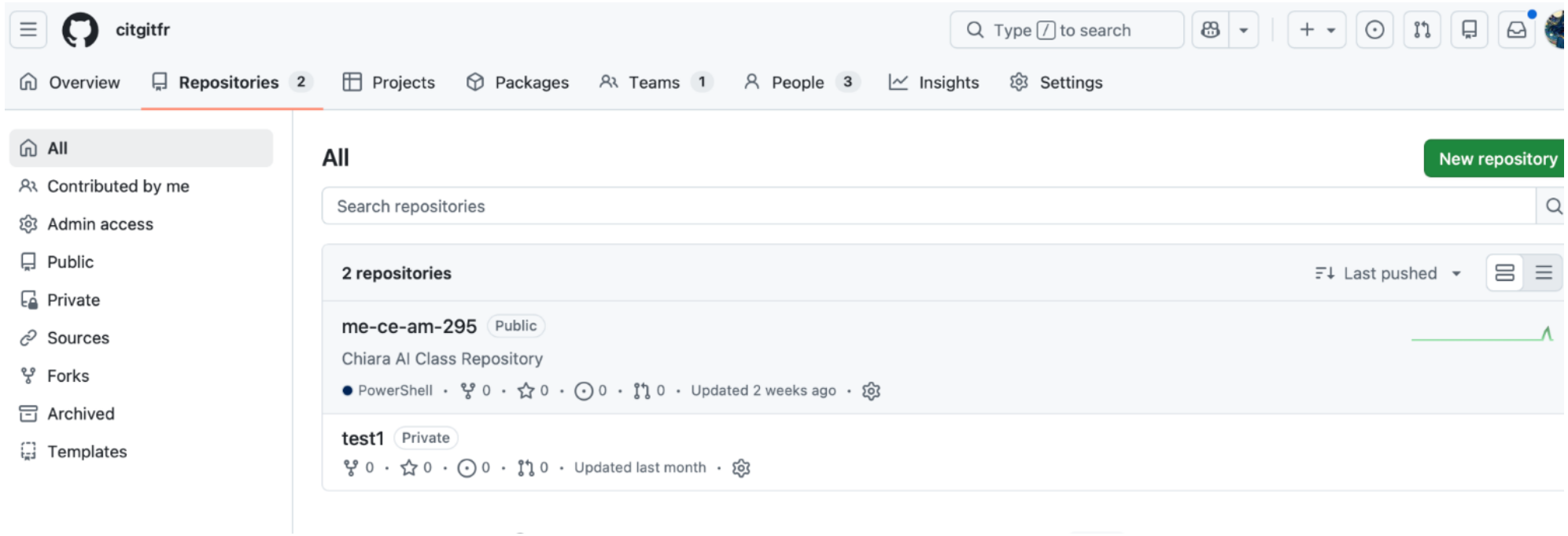
Sincerely,

Amazon WorkSpaces

LINK to download the desktop app: https://clients.amazonworkspaces.com/?

LINK to setup basic apps (Claude Code/Desktop) to AWS workspace: https://github.com/citgitfr/me-ce-am-295

What we provide for this course – 3. Github organization



- Each would create their own repo to work on the final project (+ lab report) code deliverables
- One repo per each person or team to work persistently

Assignments & Lab & Final Project

1. Weekly assignment

- Guideline: LINK
- Submit via CodeGrade

1. Biweekly Lab Report (Total 4) & Final Project Presentation+Paper

- Work in teams of 2 to develop a research project using agentic AI tools
- Use AWS workspace provided for the project
- Work on the codebase repo created under our course Github organization: https://github.com/citgitfr
- Submit via CodeGrade
- Lab report Guideline: LINK
- Final Project Guideline: LINK

From Automating Calculation to Automating Reasoning

- For 50 years, engineers have automated calculation: **FEA, CFD, numerical solvers**.
- We are now at a paradigm shift: automating **scientific reasoning and engineering design**.
- Agentic AI systems can read building codes, write and execute Python scripts, and orchestrate multi-physics simulations — **with an engineer accountable for the result.**
- Course philosophy: **"Learn by building."** Every concept introduced in lecture is implemented in lab the same week.
- By the end of the term, each student will have built an autonomous AI agent for an engineering research task.

- By end of Week 1:

  1) Explain why an LLM produces a plausible continuation, not ground truth. 2) Make a reproducible API call with structured output. 3) Write a few-shot extraction prompt that returns null when a value is absent.

Where This Course Is Going

Weeks 1–2	**Talking to models**	What an LLM is, how to prompt it, how to get structured data out.	YOU ARE HERE
Weeks 3–4	**Agents and grounding**	Planning, memory, tools; retrieval so answers cite real sources.	
Weeks 5–6	**Physics-informed surrogates**	Networks that obey PDEs, and fast stand-ins for FEA and CFD.	
Weeks 7–8	**Agents that act**	Writing and debugging simulation code; closed-loop discovery.	
Weeks 9–10	**Evidence and responsibility**	Reliability testing, reproducibility, professional practice.	

The Evolution of AI for Science

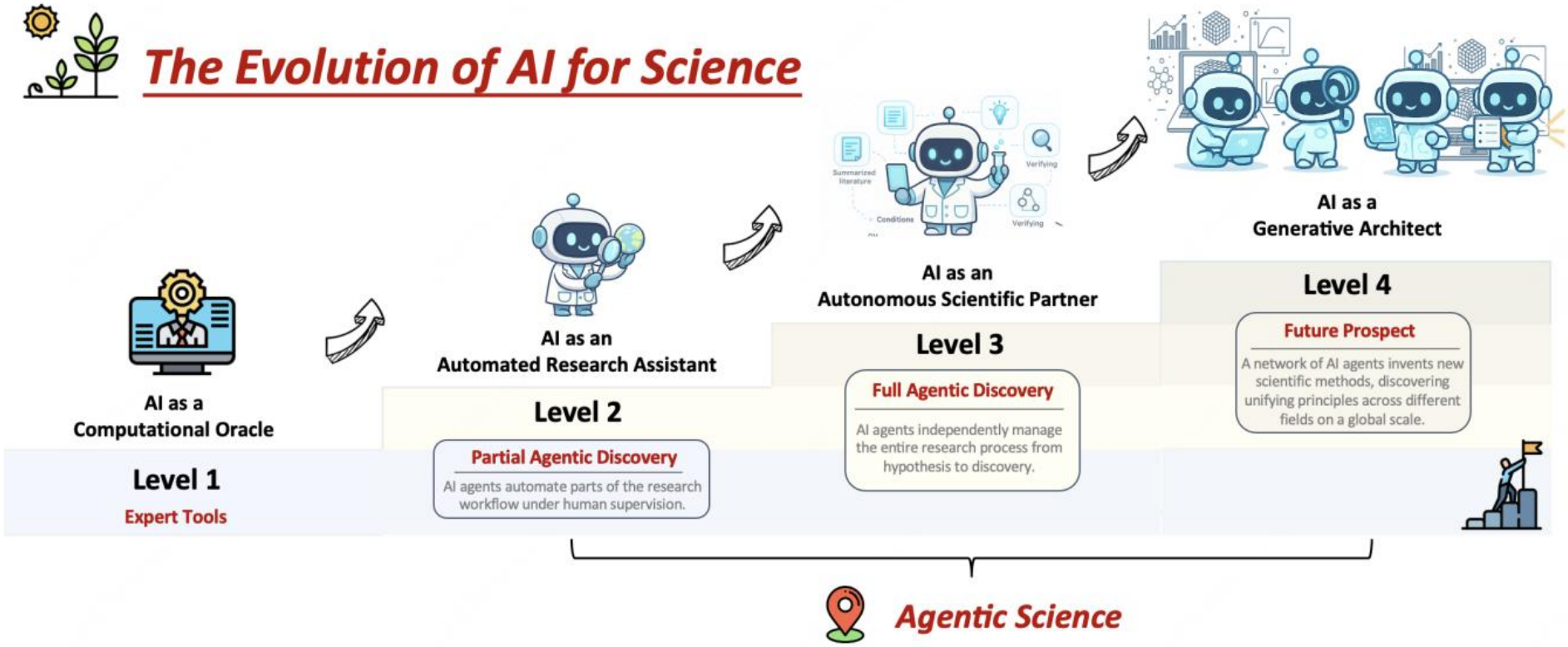


Figure 1: The Evolution of AI for Science. From computational tools to creative collaborators: the four-stage journey of AI in science. *Agentic Science is a stage within AI for Science, aligning primarily with Level 3 (Full Agentic Discovery) and building on Level 2 (Partial Agentic Discovery).*

Ref. 38, Wei, 2025, Agentic Science Survey

Sixty Years of Language Models

The lineage that leads to today — orange, on the line — and two advances outside it, in grey below, without which none of it happens.

1966
ELIZA

Pattern substitution. The illusion of understanding, with none of it.

1997
LSTM

Gated memory for sequences — but one word at a time, and slow to train.

2013
Word2Vec

Words as dense vectors. King − Man + Woman ≈ Queen.

WEEK 2 · EMBEDDINGS

2014
Attention (Bahdanau)

Attention is invented — for translation, still wrapped around recurrence.

WEEK 2 · SELF-ATTENTION

2017
Transformer

Drops the recurrence. The whole sequence at once, in parallel.

WEEK 2 · ARCHITECTURE

2020
GPT-3

Few-shot learning emerges at scale.

TODAY · LAB 1

2024–
Reasoning models

Trained to spend tokens thinking before answering.

WEEK 2 · CHAIN-OF-THOUGHT



1986

Backpropagation

Not a model — the procedure that fits every model above it. Rumelhart, Hinton & Williams.

WEEK 5 · PINNS

2012

AlexNet

Not language. GPUs + data + an ordinary architecture beat cleverness. ~60 M parameters.

NEXT SLIDE

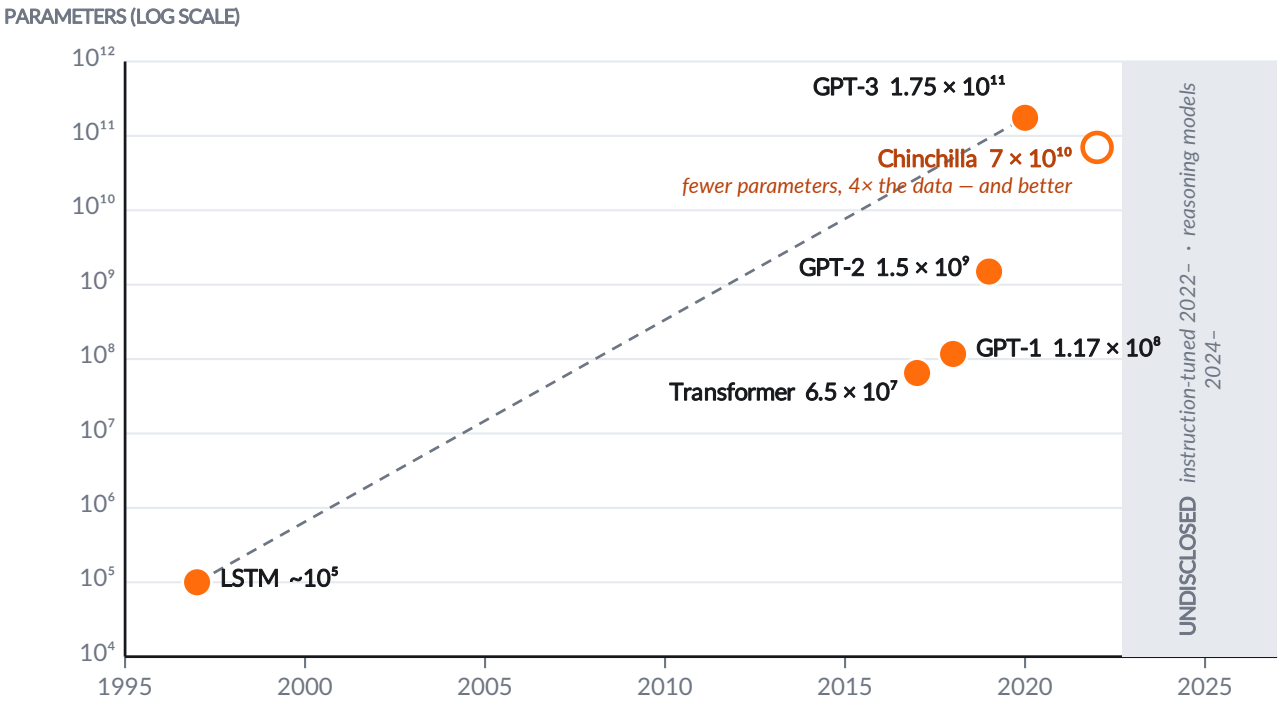
OUTSIDE THE LINEAGE. *Not language models — but the line above does not exist without them.*

SMALL CAPS UNDER EACH CARD *say where in this course the idea comes back. Nothing here is history for its own sake.*

Two of these are not language models. Backpropagation is how every model on this line is trained; AlexNet is where the field learned that compute beats cleverness. Remove either one and nothing to the right of it exists.

The Scaling Timeline

Slide 7 was the story. This is the argument — and one axis does all of the work.



Model	Parameters	What changed
1966 ELIZA	none	Pattern substitution; no model of language.
1997 LSTM	~10^{5}	Memory over sequences, one word at a time.
2017 Transformer	6.5×10^{7}	Attention without recurrence; parallel.
2018 GPT-1	1.17×10^{8}	Pre-train, then fine-tune.
2019 GPT-2	1.5×10^{9}	Coherent paragraphs; "too risky to release".
2020 GPT-3	1.75×10^{11}	Few-shot prompting, no fine-tuning.
2022 Chinchilla	7×10^{10}	2.5× fewer parameters, 4× the data — better.
2022– Instruction-tuned	undisclosed	RLHF: follows instructions, not continues.
2024– Reasoning models	undisclosed	Capability from inference compute, not size.

Six orders of magnitude in a little over twenty years. The architecture changed far less than the scale did — and the scale was not always spent well.

DISCUSSION POINT Why do the last two rows have no parameter count? And why does Chinchilla sit below the trend line yet beat the point above it?

Sources: Brown et al. 2020 (GPT-3); Hoffmann et al. 2022 (Chinchilla); Radford et al. 2018/2019; Vaswani et al. 2017 (base model). LSTM value is a typical 1990s scale, not a canonical figure.

What a Language Model Is Good At — and Is Not

RELIABLE

- Rewriting and summarising text
- Translating between formats — prose to JSON, units, notation
- Drafting code you will review
- Explaining an unfamiliar method in your own vocabulary
- Extracting structure from messy documents

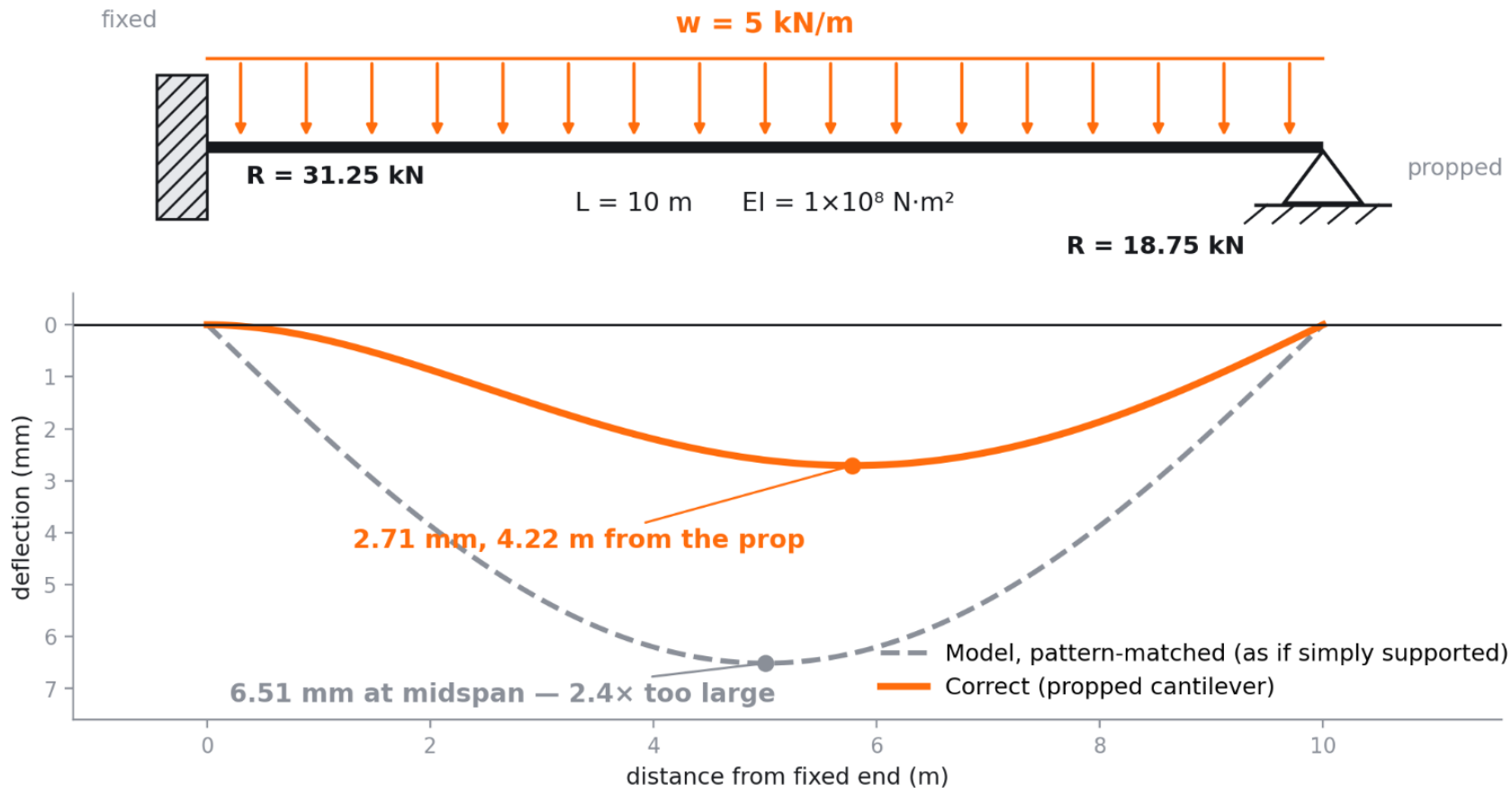
UNRELIABLE

- Arithmetic, especially multi-step
- Citations, standards clauses, property values
- Anything it was not given and cannot look up
- Knowing when it does not know
- Giving the same answer twice

LIVE DEMO — THREE MINUTES

Ask, verbatim: “A 10 m beam carries a uniform 5 kN/m. It is fixed at one end and simply supported at the other. EI = 1×10^{8} N·m². Give both support reactions, the maximum deflection, and how far from the propped end it occurs.” Then put the same question to a second tool.

The Demo Beam, Solved



Same load, same span, same EI. The bending moment is $wL^2/8$ = 62.5 kN·m in both cases — asking for the moment alone would not have exposed the mistake. **The model retrieved the nearest familiar pattern, and its confidence did not drop.**

What Still Needs You

- **It has no ground truth.** It produces the most plausible continuation of your text, which is not the same thing as the correct one.
- **It cannot look anything up** unless you give it the means — that is Week 4.
- **It cannot compute or act** unless you give it tools — that is Week 3.
- **It cannot tell you how much to trust it.** Calibration and reliability testing are Week 9.

Agentic is not autonomous

Every system you build this term keeps an engineer responsible for the result. The course is about moving that engineer from doing the work to specifying and verifying it.

THE SHAPE OF THE COURSE

Each limitation above is a week of this syllabus. That is not a coincidence — it is the design.

The Engineering Workspace

- AI libraries ship breaking changes every few months; code that runs today may not run next term.
- **The Problem:** Project A needs one version of a library, Project B needs another. A global installation breaks one of them.
- **The Solution:** Virtual environments (venv, conda) — isolated, self-contained dependency directories.
- Share a `requirements.txt` file to ensure exact version matching across different machines.

> **Core Principle:** Reproducibility is a non-negotiable requirement for scientific research.

Project A (venv)

```
langchain==0.3.x
openai==1.x
numpy==1.24.3
```

Project B (venv)

```
langchain==1.x
openai==2.x
pandas==2.1.0
```

Standard Workflow

1. Activate

↓

2. Install

↓

3. Work

↓

4. Deactivate

What Is an API? — The Restaurant Analogy



The Menu

API DOCUMENTATION

Lists the available operations (dishes) and the required inputs (ingredients/options) you must provide.



The Waiter

THE API

Takes your structured request, delivers it to the kitchen, and brings the finished response back to your table.



The Kitchen

THE SERVER / LLM

Performs the heavy lifting (computation/inference) hidden from view, returning only the final result.

Core Definition: An API (Application Programming Interface) is simply a standardized way for two pieces of software to talk to each other over the internet.

Never Hardcode Your API Key

- An API key is a password tied directly to your credit card.
- Bots scrape public GitHub repositories 24/7 looking for exposed keys.
- A leaked key can result in a **$50,000 bill** in a matter of hours.

> CHECK YOUR UNDERSTANDING
>
> The standard security practice is to store keys in a local **.env** file and exclude it from version control with **.gitignore**.

The Wrong Way (Dangerous)

```
# main.py
from anthropic import Anthropic
# Hardcoded key - WILL BE LEAKED
client = Anthropic(api_key="sk-ant-
1234567890abcdef")
```

The Right Way (.env Practice)

```
# .env (NEVER COMMIT)
ANTHROPIC_API_KEY=sk-ant-api03-1234567890abcdef
# .gitignore
.env
# main.py
from anthropic import Anthropic
client = Anthropic()
```

Recommended reading for Week 1

Required reading

1. Wei, J., Yang, Y., Zhang, X., Chen, Y., Zhuang, X., Gao, Z., Zhou, D., Wang, G., Gao, Z., Cao, J., Qiu, Z., Hu, M., Ma, C., Tang, S., He, J., Song, C., He, X., Zhang, Q., You, C., Zheng, S., Ding, N., Ouyang, W., Dong, N., Cheng, Y., Sun, S., Bai, L., & Zhou, B. (2025). *From AI for Science to Agentic Science: A Survey on Autonomous Scientific Discovery.* arXiv:2508.14111 [cs.LG], v2, 20 October 2025. https://doi.org/10.48550/arXiv.2508.14111 (preprint, not peer reviewed; 27 authors, Shanghai AI Laboratory and collaborators)
2. Schluntz, E., & Zhang, B. (2024). *Building Effective Agents.* Anthropic engineering blog, 19 December 2024. https://www.anthropic.com/engineering/building-effective-agents (industry publication, not peer reviewed)
3. Beeravolu, A. R., Kishore Babu, N., Talari, M. K., Rehman, A. U., & Srirangam, P. (2023). Influence of microstructure and mechanical properties of dissimilar rotary friction welded Inconel to stainless steel joints. *Materials*, 16(8), 3049. https://doi.org/10.3390/ma16083049 (open access, CC BY 4.0)

GNoME / A-Lab pair with critiques

4. Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. *Nature*, 624(7990), 80–85. https://doi.org/10.1038/s41586-023-06735-9 (open access)
5. Cheetham, A. K., & Seshadri, R. (2024). Artificial intelligence driving materials discovery? Perspective on the article: Scaling deep learning for materials discovery. *Chemistry of Materials*, 36(8), 3490–3495. https://doi.org/10.1021/acs.chemmater.4c00643 (open access, CC BY 4.0 — so the stub in your folder can be replaced with the full PDF without a paywall issue)
6. Szymanski, N. J., Rendy, B., Fei, Y., Kumar, R. E., He, T., Milsted, D., McDermott, M. J., Gallant, M., Cubuk, E. D., Merchant, A., Kim, H., Jain, A., Bartel, C. J., Persson, K., Zeng, Y., & Ceder, G. (2023). An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624(7990), 86–91. https://doi.org/10.1038/s41586-023-06734-w (open access). Author Correction: *Nature*, 650(8100), E1 (2026). https://doi.org/10.1038/s41586-025-09992-y
7. Leeman, J., Liu, Y., Stiles, J., Lee, S. B., Bhatt, P., Schoop, L. M., & Palgrave, R. G. (2024). Challenges in high-throughput inorganic materials prediction and autonomous synthesis. *PRX Energy*, 3(1), 011002. https://doi.org/10.1103/PRXEnergy.3.011002 (open access; preprint ChemRxiv, January 2024)